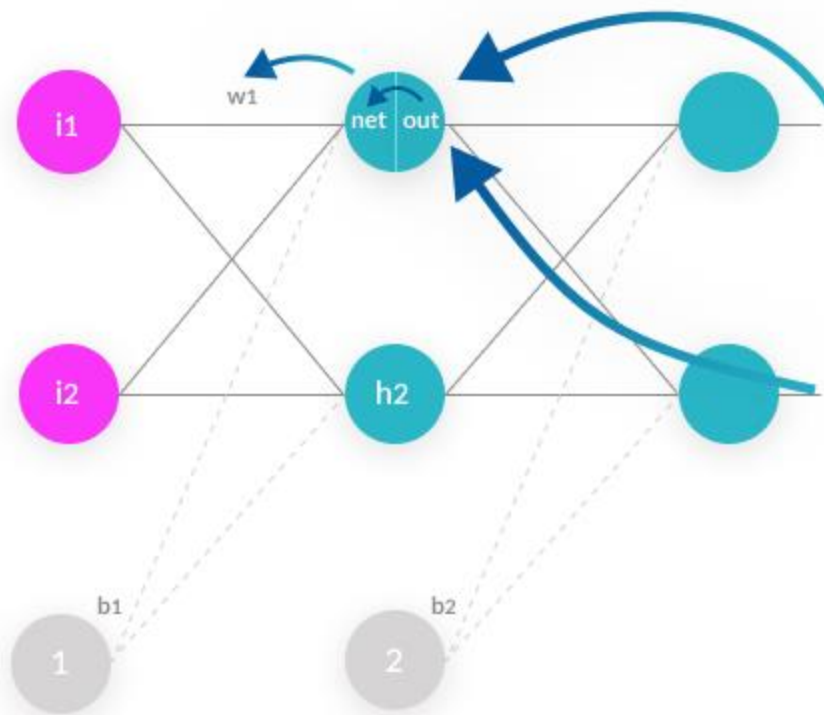


Backpropagation

What is Backpropagation?

Backpropagation is an algorithm commonly used to train neural networks. When the neural network is initialized, weights are set for its individual elements, called neurons. Inputs are loaded, they are passed through the network of neurons, and the network provides an output for each one, given the initial weights. Backpropagation helps to adjust the weights of the neurons so that the result comes closer and closer to the known true result.



It involves two main steps: a forward pass to compute the loss, and a backward pass to compute gradients and update weights.

□ **Forward Pass:**

- Compute the output of each neuron from the input layer to the output layer.
- Calculate the loss (error) at the output layer.

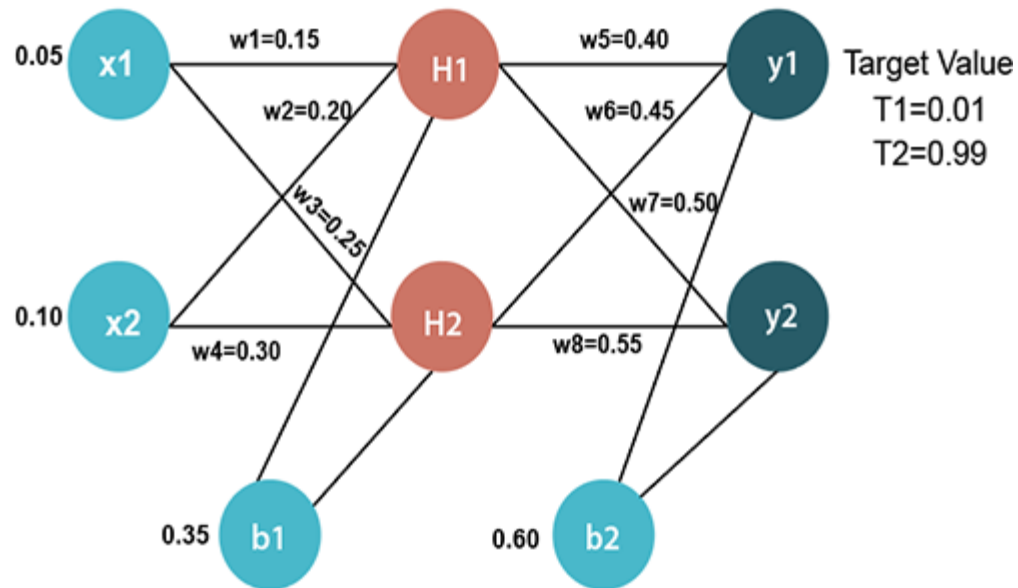
□ **Backward Pass:**

- Compute the gradient of the loss with respect to each weight using the chain rule.
- Update the weights to minimize the loss.

Components

- **Loss Function:** Measures the error between the predicted output and the actual output. Common loss functions include Mean Squared Error (MSE) and Cross-Entropy Loss.
- **Activation Functions:** Apply non-linearity to the inputs. Common activation functions include Sigmoid, ReLU, and Tanh.

Example



Input values

$X_1 = 0.05$
 $X_2 = 0.10$

Initial weight

$W_1 = 0.15$	$w_5 = 0.40$
$W_2 = 0.20$	$w_6 = 0.45$
$W_3 = 0.25$	$w_7 = 0.50$
$W_4 = 0.30$	$w_8 = 0.55$

Bias Values

$b_1 = 0.35$ $b_2 = 0.60$

Target Values

$$T1=0.01$$

$$T2=0.99$$

Now, we first calculate the values of H1 and H2 by a forward pass.

1. Forward Pass

To find the value of H1 we first multiply the input value from the weights as

$$H1=x1 \times w_1 + x2 \times w_2 + b1$$

$$H1=0.05 \times 0.15 + 0.10 \times 0.20 + 0.35$$

$$\mathbf{H1=0.3775}$$

To calculate the result of H1, we performed the sigmoid function as

$$H1_{final} = \frac{1}{1 + \frac{1}{e^{H1}}}$$

$$H1_{final} = \frac{1}{1 + \frac{1}{e^{0.3775}}}$$

$$\mathbf{H1_{final} = 0.593269992}$$

We will calculate the value of H2 in the same way as H1

$$H2=x1 \times w_3 + x2 \times w_4 + b1$$

$$H2=0.05 \times 0.25 + 0.10 \times 0.30 + 0.35$$

$$\mathbf{H2=0.3925}$$

To calculate the result of H1, we performed the sigmoid function as

$$H2_{final} = \frac{1}{1 + \frac{1}{e^{H2}}}$$

$$H2_{final} = \frac{1}{1 + \frac{1}{e^{0.3925}}}$$

$$\mathbf{H2_{final} = 0.596884378}$$

Now, we calculate the values of **y1 and y2** in the same way as we calculate the H1 and H2.

To find the value of y1, we first multiply the input value i.e., the outcome of H1 and H2 from the weights as

$$\begin{aligned} y1 &= H1 \times w_5 + H2 \times w_6 + b2 \\ y1 &= 0.593269992 \times 0.40 + 0.596884378 \times 0.45 + 0.60 \\ \mathbf{y1} &= \mathbf{1.10590597} \end{aligned}$$

To calculate the result of y1 we performed the sigmoid function as

$$\begin{aligned} y1_{final} &= \frac{1}{1 + \frac{1}{e^{y1}}} \\ y1_{final} &= \frac{1}{1 + \frac{1}{e^{1.10590597}}} \\ \mathbf{y1_{final} = 0.75136507} \end{aligned}$$

We will calculate the value of y2 in the same way as y1

$$\begin{aligned} y2 &= H1 \times w_7 + H2 \times w_8 + b2 \\ y2 &= 0.593269992 \times 0.50 + 0.596884378 \times 0.55 + 0.60 \\ y2 &= 1.2249214 \end{aligned}$$

To calculate the final result of H1, we performed the sigmoid function as

$$y2_{\text{final}} = \frac{1}{1 + \frac{1}{e^{y2}}}$$

$$y2_{\text{final}} = \frac{1}{1 + \frac{1}{e^{1.2249214}}}$$

$$y2_{\text{final}} = \mathbf{0.772928465}$$

Our target values are **0.01 and 0.99**. Our y1 and y2 value is not matched with our target values T1 and T2.

Now, we will find the **total error**, which is simply the difference between the outputs from the target outputs. The total error is calculated as

$$E_{\text{total}} = \sum \frac{1}{2} (\text{target} - \text{output})^2$$

So, the total error is

$$= \frac{1}{2} (t1 - y1_{\text{final}})^2 + \frac{1}{2} (T2 - y2_{\text{final}})^2$$

$$= \frac{1}{2} (0.01 - 0.75136507)^2 + \frac{1}{2} (0.99 - 0.772928465)^2$$

$$= 0.274811084 + 0.0235600257$$

$$\mathbf{E_{\text{total}} = 0.29837111}$$

Now, we will backpropagate this error to update the weights using a backward pass.

Backward pass at the output layer

To update the weight, we calculate the error correspond to each weight with the help of a total error. The error on weight w is calculated by differentiating total error with respect to w.

$$\text{Error}_w = \frac{\partial E_{\text{total}}}{\partial w}$$

We perform backward process so first consider the last weight w5 as

$$\text{Error}_{w5} = \frac{\partial E_{\text{total}}}{\partial w5} \dots \dots \dots (1)$$

$$E_{\text{total}} = \frac{1}{2} (T1 - y1_{\text{final}})^2 + \frac{1}{2} (T2 - y2_{\text{final}})^2 \dots \dots \dots (2)$$

From equation two, it is clear that we cannot partially differentiate it with respect to w5 because there is no any w5. We split equation one into multiple terms so that we can easily differentiate it with respect to w5 as

$$\frac{\partial E_{\text{total}}}{\partial w5} = \frac{\partial E_{\text{total}}}{\partial y1_{\text{final}}} \times \frac{\partial y1_{\text{final}}}{\partial y1} \times \frac{\partial y1}{\partial w5} \dots \dots \dots (3)$$

Now, we calculate each term one by one to differentiate E_{total} with respect to w5 as

$$\begin{aligned} \frac{\partial E_{\text{total}}}{\partial y1_{\text{final}}} &= \frac{\partial (\frac{1}{2} (T1 - y1_{\text{final}})^2 + \frac{1}{2} (T2 - y2_{\text{final}})^2)}{\partial y1_{\text{final}}} \\ &= 2 \times \frac{1}{2} \times (T1 - y1_{\text{final}})^{2-1} \times (-1) + 0 \\ &= -(T1 - y1_{\text{final}}) \\ &= -(0.01 - 0.75136507) \end{aligned}$$

$$\frac{\partial E_{\text{total}}}{\partial y1_{\text{final}}} = 0.74136507 \dots \dots \dots (4)$$

$$y1_{\text{final}} = \frac{1}{1 + e^{-y1}} \dots \dots \dots (5)$$

$$\begin{aligned} \frac{\partial y1_{\text{final}}}{\partial y1} &= \frac{\partial (\frac{1}{1 + e^{-y1}})}{\partial y1} \\ &= \frac{e^{-y1}}{(1 + e^{-y1})^2} \\ &= e^{-y1} \times (y1_{\text{final}})^2 \dots \dots \dots (6) \end{aligned}$$

$$y1_{\text{final}} = \frac{1}{1 + e^{-y1}}$$

$$e^{-y1} = \frac{1 - y1_{\text{final}}}{y1_{\text{final}}} \dots \dots \dots (7)$$

Putting the value of e^{-y} in equation (5)

$$\begin{aligned}
 &= \frac{1 - y1_{final}}{y1_{final}} \times (y1_{final})^2 \\
 &= y1_{final} \times (1 - y1_{final}) \\
 &= 0.75136507 \times (1 - 0.75136507)
 \end{aligned}$$

$$\frac{\partial y1_{final}}{\partial y1} = 0.186815602 \dots \dots \dots (8)$$

$$y1 = H1_{final} \times w5 + H2_{final} \times w6 + b2 \dots \dots \dots (9)$$

$$\begin{aligned}
 \frac{\partial y1}{\partial w5} &= \frac{\partial (H1_{final} \times w5 + H2_{final} \times w6 + b2)}{\partial w5} \\
 &= H1_{final}
 \end{aligned}$$

$$\frac{\partial y1}{\partial w5} = 0.596884378 \dots \dots \dots (10)$$

So, we put the values of $\frac{\partial E_{total}}{\partial y1_{final}}$, $\frac{\partial y1_{final}}{\partial y1}$, and $\frac{\partial y1}{\partial w5}$ in equation no (3) to find the final result.

$$\begin{aligned}
 \frac{\partial E_{total}}{\partial w5} &= \frac{\partial E_{total}}{\partial y1_{final}} \times \frac{\partial y1_{final}}{\partial y1} \times \frac{\partial y1}{\partial w5} \\
 &= 0.74136507 \times 0.186815602 \times 0.593269992 \\
 \text{Error}_{w5} &= \frac{\partial E_{total}}{\partial w5} = 0.0821670407 \dots \dots \dots (11)
 \end{aligned}$$

Now, we will calculate the updated weight $w5_{new}$ with the help of the following formula

$$\begin{aligned}
 w5_{new} &= w5 - \eta \times \frac{\partial E_{total}}{\partial w5} \text{ Here, } \eta = \text{learning rate} = 0.5 \\
 &= 0.4 - 0.5 \times 0.0821670407 \\
 w5_{new} &= 0.35891648 \dots \dots \dots (12)
 \end{aligned}$$

In the same way, we calculate $w6_{new}$, $w7_{new}$, and $w8_{new}$ and this will give us the following values

$$\begin{aligned}
 w5_{new} &= 0.35891648 \\
 w6_{new} &= 408666186
 \end{aligned}$$

$$\begin{aligned}w7_{\text{new}} &= 0.511301270 \\ w8_{\text{new}} &= 0.561370121\end{aligned}$$

Backward pass at Hidden layer

Now, we will backpropagate to our hidden layer and update the weight $w1$, $w2$, $w3$, and $w4$ as we have done with $w5$, $w6$, $w7$, and $w8$ weights.

We will calculate the error at $w1$ as

$$\begin{aligned}\text{Error}_{w1} &= \frac{\partial E_{\text{total}}}{\partial w1} \\ E_{\text{total}} &= \frac{1}{2}(T1 - y1_{\text{final}})^2 + \frac{1}{2}(T2 - y2_{\text{final}})^2\end{aligned}$$

From equation (2), it is clear that we cannot partially differentiate it with respect to $w1$ because there is no any $w1$. We split equation (1) into multiple terms so that we can easily differentiate it with respect to $w1$ as

$$\frac{\partial E_{\text{total}}}{\partial w1} = \frac{\partial E_{\text{total}}}{\partial H1_{\text{final}}} \times \frac{\partial H1_{\text{final}}}{\partial H1} \times \frac{\partial H1}{\partial w1} \dots \dots \dots (13)$$

Now, we calculate each term one by one to differentiate E_{total} with respect to $w1$ as

$$\frac{\partial E_{\text{total}}}{\partial H1_{\text{final}}} = \frac{\partial (\frac{1}{2}(T1 - y1_{\text{final}})^2 + \frac{1}{2}(T2 - y2_{\text{final}})^2)}{\partial H1} \dots \dots \dots (14)$$

We again split this because there is no any $H1_{\text{final}}$ term in E_{total} as

$$\frac{\partial E_{\text{total}}}{\partial H1_{\text{final}}} = \frac{\partial E_1}{\partial H1_{\text{final}}} + \frac{\partial E_2}{\partial H1_{\text{final}}} \dots \dots \dots (15)$$

$\frac{\partial E_1}{\partial H1_{\text{final}}}$ and $\frac{\partial E_2}{\partial H1_{\text{final}}}$ will again split because in $E1$ and $E2$ there is no $H1$ term. Splitting is done as

$$\frac{\partial E_1}{\partial H1_{final}} = \frac{\partial E_1}{\partial y1} \times \frac{\partial y1}{\partial H1_{final}} \dots \dots \dots (16)$$

$$\frac{\partial E_2}{\partial H1_{final}} = \frac{\partial E_2}{\partial y2} \times \frac{\partial y2}{\partial H1_{final}} \dots \dots \dots (17)$$

We again Split both $\frac{\partial E_1}{\partial y1}$ and $\frac{\partial E_2}{\partial y2}$ because there is no any y1 and y2 term in E1 and E2. We split it as

$$\frac{\partial E_1}{\partial y1} = \frac{\partial E_1}{\partial y1_{final}} \times \frac{\partial y1_{final}}{\partial y1} \dots \dots \dots (18)$$

$$\frac{\partial E_2}{\partial y2} = \frac{\partial E_2}{\partial y2_{final}} \times \frac{\partial y2_{final}}{\partial y2} \dots \dots \dots (19)$$

Now, we find the value of $\frac{\partial E_1}{\partial y1}$ and $\frac{\partial E_2}{\partial y2}$ by putting values in equation (18) and (19) as

From equation (18)

$$\begin{aligned} \frac{\partial E_1}{\partial y1} &= \frac{\partial E_1}{\partial y1_{final}} \times \frac{\partial y1_{final}}{\partial y1} \\ &= \frac{\partial (\frac{1}{2}(T1 - y1_{final})^2)}{\partial y1_{final}} \times \frac{\partial y1_{final}}{\partial y1} \\ &= 2 \times \frac{1}{2}(T1 - y1_{final}) \times (-1) \times \frac{\partial y1_{final}}{\partial y1} \end{aligned}$$

From equation (8)

$$\begin{aligned} &= 2 \times \frac{1}{2}(0.01 - 0.75136507) \times (-1) \times 0.186815602 \\ \frac{\partial E_1}{\partial y1} &= \mathbf{0.138498562} \dots \dots \dots (20) \end{aligned}$$

From equation (19)

$$\begin{aligned}
\frac{\partial E_2}{\partial y_2} &= \frac{\partial E_2}{\partial y_{2_{\text{final}}}} \times \frac{\partial y_{2_{\text{final}}}}{\partial y_2} \\
&= \frac{\partial(\frac{1}{2}(T_2 - y_{2_{\text{final}}})^2)}{\partial y_{2_{\text{final}}}} \times \frac{\partial y_{2_{\text{final}}}}{\partial y_2} \\
&= 2 \times \frac{1}{2}(T_2 - y_{2_{\text{final}}}) \times (-1) \times \frac{\partial y_{2_{\text{final}}}}{\partial y_2} \dots \dots \dots (21)
\end{aligned}$$

$$y_{2_{\text{final}}} = \frac{1}{1 + e^{-y_2}} \dots \dots \dots (22)$$

$$\begin{aligned}
\frac{\partial y_{2_{\text{final}}}}{\partial y_2} &= \frac{\partial(\frac{1}{1 + e^{-y_2}})}{\partial y_2} \\
&= \frac{e^{-y_2}}{(1 + e^{-y_2})^2} \\
&= e^{-y_2} \times (y_{2_{\text{final}}})^2 \dots \dots \dots (23)
\end{aligned}$$

$$y_{2_{\text{final}}} = \frac{1}{1 + e^{-y_2}}$$

$$e^{-y_2} = \frac{1 - y_{2_{\text{final}}}}{y_{2_{\text{final}}}} \dots \dots \dots (24)$$

Putting the value of e^{-y_2} in equation (23)

$$\begin{aligned}
&= \frac{1 - y_{2_{\text{final}}}}{y_{2_{\text{final}}}} \times (y_{2_{\text{final}}})^2 \\
&= y_{2_{\text{final}}} \times (1 - y_{2_{\text{final}}}) \\
&= 0.772928465 \times (1 - 0.772928465) \\
\frac{\partial y_{2_{\text{final}}}}{\partial y_2} &= \mathbf{0.175510053} \dots \dots \dots (25)
\end{aligned}$$

From equation (21)

$$= 2 \times \frac{1}{2} (0.99 - 0.772928465) \times (-1) \times 0.175510053$$

$$\frac{\partial E_1}{\partial y_1} = -0.0380982366126414 \dots \dots (26)$$

Now from equation (16) and (17)

$$\begin{aligned} \frac{\partial E_1}{\partial H1_{final}} &= \frac{\partial E_1}{\partial y_1} \times \frac{\partial y_1}{\partial H1_{final}} \\ &= 0.138498562 \times \frac{\partial (H1_{final} \times w_5 + H2_{final} \times w_6 + b2)}{\partial H1_{final}} \\ &= 0.138498562 \times \frac{\partial (H1_{final} \times w_5 + H2_{final} \times w_6 + b2)}{\partial H1_{final}} \\ &= 0.138498562 \times w_5 \\ &= 0.138498562 \times 0.40 \end{aligned}$$

$$\frac{\partial E_1}{\partial H1_{final}} = 0.0553994248 \dots \dots (27)$$

$$\begin{aligned} \frac{\partial E_2}{\partial H1_{final}} &= \frac{\partial E_2}{\partial y_2} \times \frac{\partial y_2}{\partial H1_{final}} \\ &= -0.0380982366126414 \times \frac{\partial (H1_{final} \times w_7 + H2_{final} \times w_8 + b2)}{\partial H1_{final}} \\ &= -0.0380982366126414 \times w_7 \\ &= -0.0380982366126414 \times 0.50 \end{aligned}$$

$$\frac{\partial E_2}{\partial H1_{final}} = -0.0190491183063207 \dots \dots (28)$$

Put the value of $\frac{\partial E_1}{\partial H1_{final}}$ and $\frac{\partial E_2}{\partial H1_{final}}$ in equation (15) as

$$\frac{\partial E_{\text{total}}}{\partial H1_{\text{final}}} = \frac{\partial E_1}{\partial H1_{\text{final}}} + \frac{\partial E_2}{\partial H1_{\text{final}}}$$

$$= 0.0553994248 + (-0.0190491183063207)$$

$$\frac{\partial E_{\text{total}}}{\partial H1_{\text{final}}} = \mathbf{0.0364908241736793 \dots \dots \dots (29)}$$

We have $\frac{\partial E_{\text{total}}}{\partial H1_{\text{final}}}$, we need to figure out $\frac{\partial H1_{\text{final}}}{\partial H1}$, $\frac{\partial H1}{\partial w1}$ as

$$\begin{aligned} \frac{\partial H1_{\text{final}}}{\partial H1} &= \frac{\partial \left(\frac{1}{1 + e^{-H1}} \right)}{\partial H1} \\ &= \frac{e^{-H1}}{(1 + e^{-H1})^2} \\ e^{-H1} \times (H1_{\text{final}})^2 \dots \dots \dots (30) \end{aligned}$$

$$H1_{\text{final}} = \frac{1}{1 + e^{-H1}}$$

$$e^{-H1} = \frac{1 - H1_{\text{final}}}{H1_{\text{final}}} \dots \dots \dots (31)$$

Putting the value of e^{-H1} in equation (30)

$$\begin{aligned} &= \frac{1 - H1_{\text{final}}}{H1_{\text{final}}} \times (H1_{\text{final}})^2 \\ &= H1_{\text{final}} \times (1 - H1_{\text{final}}) \\ &= 0.593269992 \times (1 - 0.593269992) \\ \frac{\partial H1_{\text{final}}}{\partial H1} &= \mathbf{0.2413007085923199} \end{aligned}$$

We calculate the partial derivative of the total net input to H1 with respect to w1 the same as we did for the output neuron:

$$H1 = H1_{final} \times w5 + H2_{final} \times w6 + b2 \dots \dots \dots (32)$$

$$\frac{\partial y1}{\partial w1} = \frac{\partial(x1 \times w1 + x2 \times w3 + b1 \times 1)}{\partial w1}$$

$$= x1$$

$$\frac{\partial H1}{\partial w1} = 0.05 \dots \dots \dots (33)$$

So, we put the values of $\frac{\partial E_{total}}{\partial H1_{final}}$, $\frac{\partial H1_{final}}{\partial H1}$, and $\frac{\partial H1}{\partial w1}$ in equation (13) to find the final result.

$$\frac{\partial E_{total}}{\partial w1} = \frac{\partial E_{total}}{\partial H1_{final}} \times \frac{\partial H1_{final}}{\partial H1} \times \frac{\partial H1}{\partial w1}$$

$$= 0.0364908241736793 \times 0.2413007085923199 \times 0.05$$

$$\text{Error}_{w1} = \frac{\partial E_{total}}{\partial w1} = 0.000438568 \dots \dots \dots (34)$$

Now, we will calculate the updated weight $w1_{new}$ with the help of the following formula

$$w1_{new} = w1 - \eta \times \frac{\partial E_{total}}{\partial w1} \text{ Here } \eta = \text{learning rate} = 0.5$$

$$= 0.15 - 0.5 \times 0.000438568$$

$$w1_{new} = 0.149780716 \dots \dots \dots (35)$$

In the same way, we calculate $w2_{new}$, $w3_{new}$, and $w4$ and this will give us the following values

$$w1_{new} = 0.149780716$$

$$w2_{new} = 0.19956143$$

$$w3_{new} = 0.24975114$$

$$w4_{new} = 0.29950229$$

We have updated all the weights. We found the error 0.298371109 on the network when we fed forward the 0.05 and 0.1 inputs. In the first round of Backpropagation, the total error is down to 0.291027924. After repeating this process 10,000, the total error is down to 0.0000351085. At this point, the outputs neurons generate 0.159121960 and 0.984065734 i.e., nearby our target value when we feed forward the 0.05 and 0.1.

